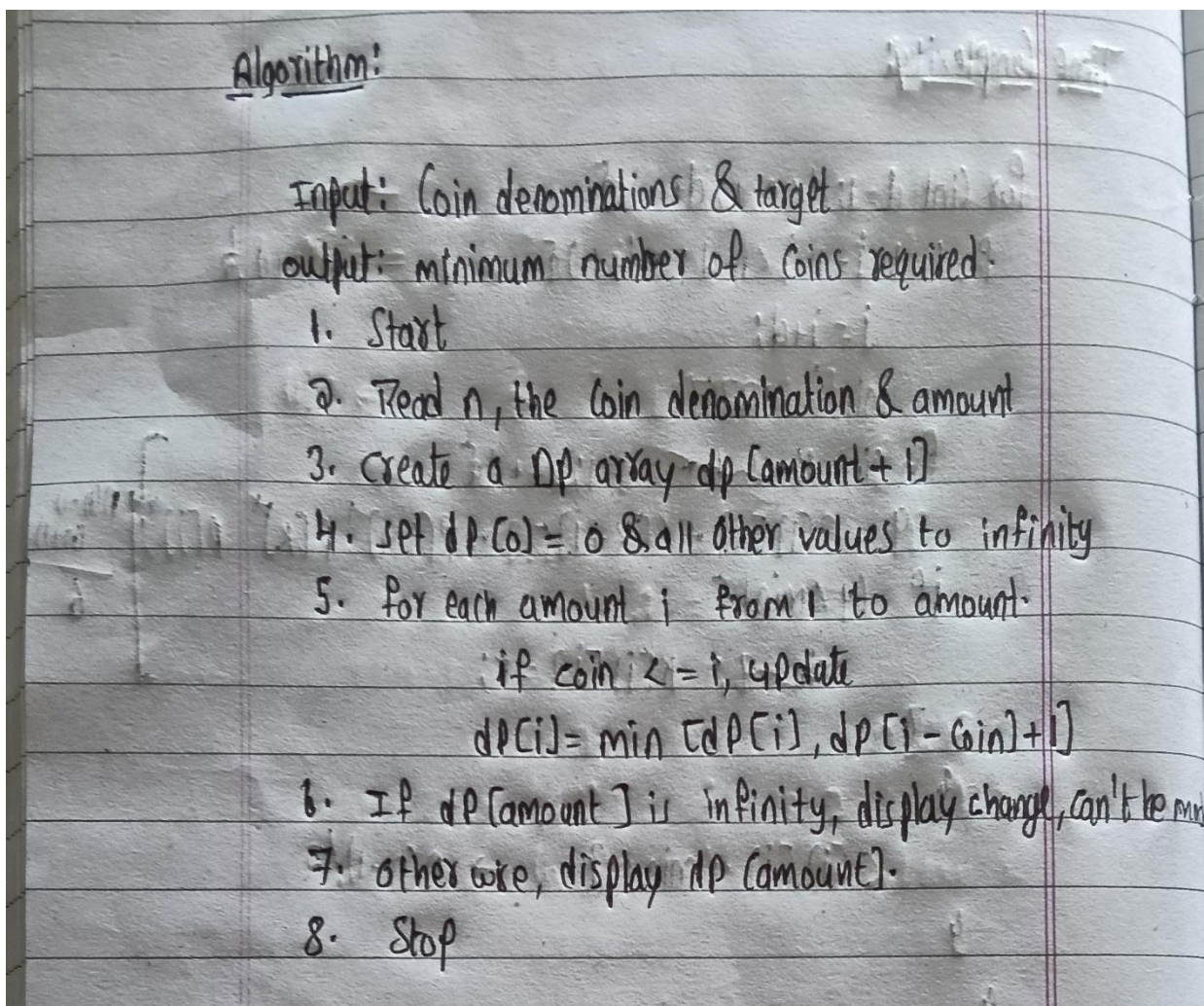


Practical - 7

Aim : To implement the Making Change problem using Dynamic Programming and find the minimum number of coins required to make a given amount.

Algorithm :



Program / Code :

```
#include <iostream>
using namespace std;
int main() {    int n,
amount;
    cout << "Enter number of coins: ";
cin >> n;    int coins[n];    cout <<
"Enter the coins: ";
    for (int i = 0; i < n; i++) {

        cin >> coins[i];
    }
    cout << "Enter amount: ";
cin >> amount;    int
dp[amount + 1];
    dp[0] = 0;
    for (int i = 1; i <= amount; i++) {
dp[i] = 9999;
    }
    for (int i = 1; i <= amount; i++) {
for (int j = 0; j < n; j++) {        if
(coins[j] <= i) {
            int combined_coins = dp[i - coins[j]] + 1;
if (combined_coins < dp[i]) {
                dp[i] = combined_coins;
            }
        }
    }
}
    cout << "Minimum number of coins = " << dp[amount] << endl;
return 0;
}
```

Output :

```
Enter number of coins: 4
Enter the coins: 2
6
4
3
Enter amount: 12
Minimum number of coins = 2
```

Time Complexity :

Time Complexity:

n = no. of coin denominations.
 A = target Amount

The algorithm has two nested loops.

for (int i=1; i<=amount; i++) $\rightarrow$ A times
for (int j=0; j<n; j++) $\rightarrow$ n times

if (coins[j] <= i) $\rightarrow$ $O(1)$

$dp[i] = \min(dp[i], dp[i - \text{coins}[j]] + 1);$

$\therefore$

⇒ The outer loop executes A times & the inner loop executes n times for every amount.

therefore : $T(n, A) = A \times n$

$$T(n, A) = O(nA)$$

Best case:

Even if the first coin gives the $\min^m$ the inner loop still checks all n coins for every amount.

$$T(n, A) = nA$$

$$\text{Best case} = O(nA)$$

Average case:

on average the algorithm still calculates all A amounts coins denominations

$$T(n, A) = nA$$

$$\text{Avg case} = O(nA)$$

Worst case:

In worst case, the algorithm also calculates $n \times A$ DP entries

$$T(n, A) = nA$$

$$\text{worst case} = O(nA)$$

Conclusion :

Thus, the Making Change problem was successfully implemented using Dynamic Programming, and the minimum number of coins required to form the given amount was determined efficiently

